

# Nathaniel Rogers

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Dear Hiring Manager,

I am applying to the Entry-Level Systems Engineer position at Boeing. With a background in astrophysics, coupled with hands-on experience in aerospace project management and mission design, I am excited about the opportunity to contribute to the innovative work being done at Boeing.

My academic background is in astrophysics; I have a Bachelor of Science in Earth and Space Exploration, with a emphasis in Astrophysics from Arizona State University. I actively sought out opportunities to apply my academic knowledge to real-world scenarios. Engaging in projects such as SpaceWorks and my senior capstone, provided me with experience in team-based collaboration and model-based systems engineering in Cameo. Systems Engineering for Exploration gave me direct experience working in a systems engineering team; we created requirements, designed a CubeSat mission, and presented design reviews. I supplemented my degree path with coursework in project management.

For my senior capstone project, my team investigated the application of model based systems engineering software and methodologies in the context of cubesat missions. My group was provided the thermal telemetry data from the LunaH mapper, a mission flown by ASU. I used CATIA Magic System of Systems Architect to model the structure of the LunaH satellite in SysML. I built a rough model of the thermal performance of the satellite. We used this thermal model in addition to a thermal anomaly investigate under what conditions thermal anomalies could occur. This project lays the foundation for more extensive verification of CubeSat thermal performance before launch.

During my time in the SpaceWorks program, I was involved in a semester-long project where I played a pivotal role in designing the mechanical housing and internal structure of an infrared instrument. Assuming the role of project manager, I orchestrated team efforts, established schedules, and ensured adherence to project milestones. The successful assembly and testing of the instrument in a vacuum environment underscored our team's commitment to precision and quality, resulting in a thermal accuracy of 2.5 degrees. These experiences have fortified my proficiency in aerospace project design, management, and functional analysis.

My professional experience outside of engineering-related roles includes time at Wells Fargo Bank validating adherence to regulatory requirements for mortgages, and time managing a team of over 80 employees at a water park. I am confident that my academic background and practical experience make me a strong candidate for the Entry-Level Systems Engineer position at Boeing. I have real, hands-on experience writing technical requirements, using MBSE programs, and agile systems engineering. I am enthusiastic about the prospect of discussing how my skills and experiences can contribute to the success of your team. Thank you for considering my application. I look forward to the possibility of further discussing how I can contribute to Boeing's continued success.

Sincerely,

Nathaniel Rogers